GGR376 Project Proposal: Spatial Analysis of Diabetes in the Greater Toronto Area

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1. RESEARCH OBJECTIVES

This study aims to examine the spatial relationship between access to recreational facilities and diabetes prevalence among individuals aged 12 and above across census tracts in the City of Toronto, Peel Region, and York Region. By focusing on neighbourhood-level patterns, the research investigates how variations in access to recreational infrastructure may correspond with geographic differences in diabetes prevalence across the study area.

The study will first explore the spatial distribution of both diabetes prevalence and recreational facilities to identify geographic patterns and potential inequalities in access to opportunities for physical activity. Mapping these variables allows for an initial assessment of whether areas with higher diabetes prevalence appear to coincide with areas that have fewer recreational resources. Finally, the study also examines whether diabetes prevalence exhibits statistically significant spatial clustering across census tracts. The study also evaluates whether variations in recreational facility accessibility are associated with differences in diabetes prevalence across the region. By examining this relationship, the research contributes to a broader understanding of how environmental conditions may relate to population health outcomes in the Greater Toronto Area.

2. DATA SOURCES AND SPATIAL UNITS

Data will be primarily sourced from Statistics Canada for its accessibility, comprehensiveness, and quality. The primary health data is sourced from the Canadian Community Health Survey administered by Statistics Canada. The dataset includes the raw number of people over 12 years old who have diabetes (Type I and II) as well as the total population surveyed in each census

tract in the Greater Toronto Area for 2025. The second dataset for recreational spaces is located in the Open Database of Recreational and Sport Facilities (ODRSF), also maintained by Statistics Canada. The database compiles recreational spaces sourced from provincial, territorial, and municipal governments across Canada and is up to date as of 2021. The main variables the team will be using in this dataset are the longitude, latitude, and type of recreational facility of each recreational space. For this study, the team is only considering facility types that are conducive to physical fitness and easily accessible (i.e. excluding arenas, casinos, marinas, race tracks, splash pads, and stadiums). Additional datasets concerning the locations of restaurants, schools, supermarkets, and bus stations may be considered in the future.

3. PRELIMINARY ANALYSIS

Due to the presence of substantial missing values in the diabetes census tract data for the Halton and Durham regions, these regions were excluded from our overall analysis. As such, references to the Greater Toronto Area (GTA) are only applicable to the City of Toronto, Peel, and York Municipal Regions and any findings should also be assumed to be within this geographic scope. The mapped polygons may appear to extend beyond the conventional GTA land boundary because the dataset incorporates portions of water features (lakes) that fall within municipal jurisdictional limits. Below is the map of the selected study area.

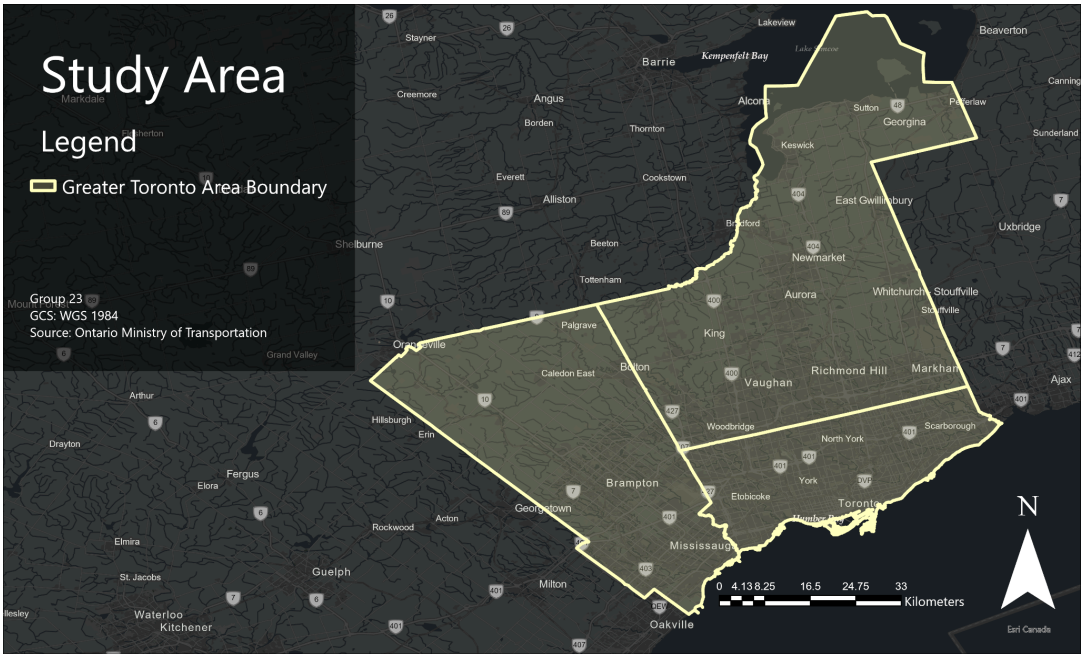

Figure 1 (above): The study area of the Peel, York, and Toronto Regional Municipalities.

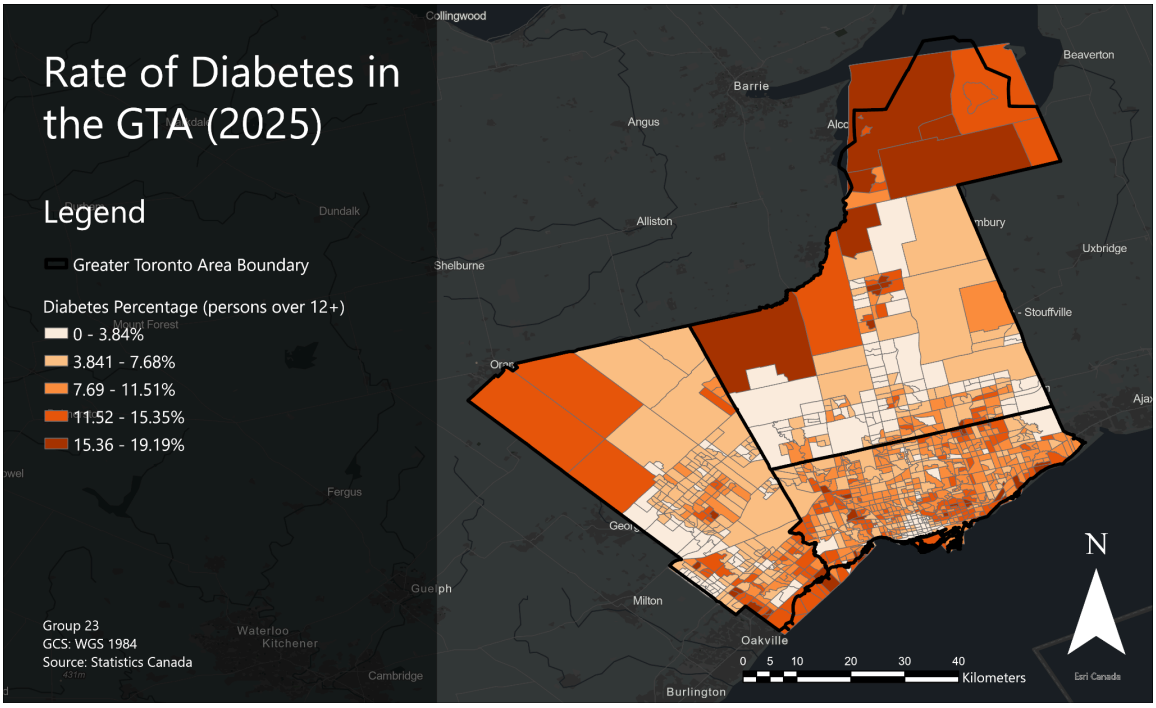

Figure 2 (above): Percentage of diabetes for persons above 12 years old in the Peel, York, and Toronto Regional Municipalities by census tract

Figure 1 shows the study area, which includes Toronto, York, and Peel, while Figure 2 is a choropleth map of diabetes rates among persons aged 12 and older by census tract across these regional municipalities. The data drawn from the 2025 Canadian Community Health Survey. For each census tract, the rate was calculated by dividing the crude number of surveyed respondents with diabetes (ages 12+) by the total surveyed respondents in the same age group. Normalizing accounts for population size differences and allows comparisons across census tracts. Figure 2 also shows that Toronto contains far more census tracts than York and Peel, with several scattered hotspots and coldspots visible across the study area.

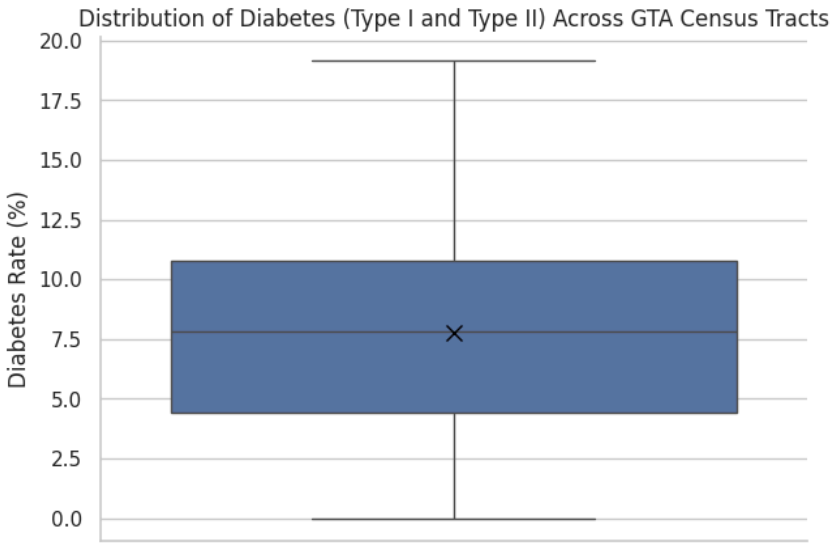

Figure 3 (above): Box plot of the diabetes prevalence by census tract for Peel, York, and Toronto Regional Municipalities in total.

Figure 3 illustrates the distribution of diabetes prevalence across the Greater Toronto Area census tracts. The values spread from 0.0% to 19.1%, indicating that while some areas show comparatively lower diabetes rates, there are other areas which tend to have very high prevalence. The median is about 7.8%, and interquartile range goes from approximately 4.4% to 10.7%, meaning that the middle 50% (25th percentile to 75th percentile) of the census tracts

have diabetes rates falling between this range. The longer upper whisker highlights the presence of census tracts with a higher diabetes rate compared to others. The mean is 7.7%, which is similar to the median.

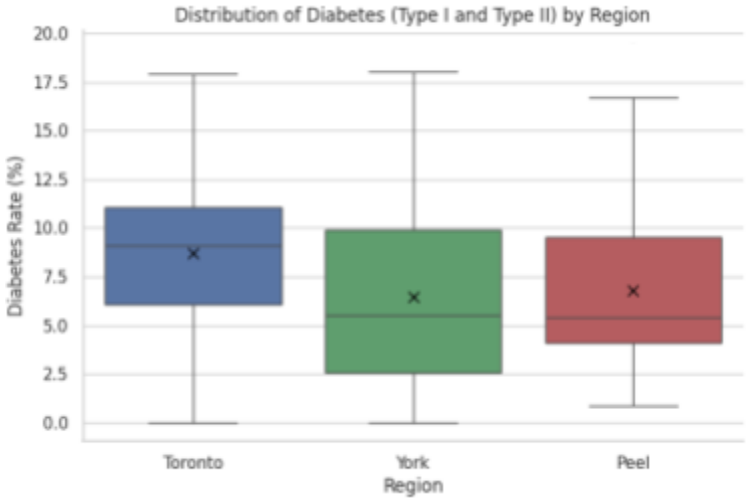

Figure 4: Box plot of the distribution for diabetes prevalence by census tract for Toronto, York, and Peel Municipal Regions, individually by Region.

Figure 4 compares the distribution of diabetes rates across Toronto, York and Peel regions for comparison. Toronto has a near-normal distribution of diabetes rates across its census tracts, while York and Peel are both positively skewed, indicated by the lengths of their whiskers and locations of their means relative to the medians. York has the largest interquartile range, followed by Peel and then Toronto, indicating York has a more variability of diabetes rates across its census tracts than Toronto and Peel. In addition, Toronto has a substantially larger median than York and Peel, indicating a higher prevalence of diabetes overall.

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Summary output by Region
<class 'pandas.core.frame.DataFrame'>
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	CDName	n_tracts	mean_percentage	max_percentage	median_percentage
0	Peel	282	6.800154	19.1892	5.39735
1	Toronto	585	8.679399	17.9148	9.11230
2	York	198	6.437566	18.0485	5.54165

Table 1: Grouped summary for Toronto, York, and Peel Regions.

Table 1 summarizes the diabetes rate across Peel, Toronto and the York regions. Toronto has the highest value for the mean diabetes percentage of 8.6%, with Peel and York having 6.8% and 6.4%, respectively. Similarly, Toronto has the highest median percentage value of 9.1% compared to 5.3% and 5.5% of Peel and York regions. These high values of mean and median explain that Toronto census tracts tend to have a higher prevalence of diabetes as compared to the other regions. On the other hand, Peel has the highest maximum diabetes percentage value at 19.1%, with York having 18.0%, and Toronto having the lowest at 17.9%. While Toronto has the highest average and the largest number of census tracts of around 585, compared to 282 in the Peel region and 198 in York, the values of all the regions tend to be very similar and display higher values.

4. PROJECT METHODOLOGY

The methodology combines exploratory spatial data analysis and spatial statistical methods to assess whether recreational facility accessibility may be associated with diabetes prevalence.

✓ Data Preparation

- Import datasets using GeoPandas and Pandas
- Load census tract shapefiles for Toronto, Peel, and York
- Convert recreational facility coordinates (longitude, latitude) into spatial points
- Filter recreational facilities to include only those promoting physical activity
- Perform a spatial join to associate recreational facilities with census tracts
- Compute recreational facility density per census tract

Output:

- Final GeoDataFrame containing:
  - census tract geometry
  - diabetes prevalence
  - recreational facility count or density

✓ Exploratory Spatial Data Analysis (ESDA)

- Descriptive statistics of diabetes prevalence
- Choropleth maps of diabetes rates
- Visualization of recreational facility distribution
- Scatter plots between recreational access and diabetes prevalence

○ Spatial Weights Matrix Construction

- Construct a Queen contiguity spatial weights matrix, where census tracts sharing boundaries or vertices are considered neighbors.

Libraries**: PySAL (specifically libpysal)**

○ Spatial Lag Calculation

The spatial lag is computed as Wy where:

- W = spatial weights matrix
- y = diabetes prevalence variable
- Wy = spatial lag of diabetes prevalence

○ Global Spatial Autocorrelation

- Compute Moran's I statistic
  - Moran's I > 0 → clustering
  - Moran's I ≈ 0 → random pattern
  - Moran's I < 0 → spatial dispersion

Libraries: **PySAL**

○ Local Spatial Autocorrelation (LISA)

- Compute Local Indicators of Spatial Association (LISA)

  Output:
  - High–High clusters (hotspots)
  - Low–Low clusters (coldspots)
  - High–Low and Low–High spatial outliers

○ Spatial Regression Analysis

If spatial dependence exists → run Spatial Lag Model (SLM) or Spatial Error Model (SEM)

○ Interpretation and Validation

Results will be interpreted in a spatial context by evaluating model coefficients and significance, along with spatial clusters and residual spatial autocorrelation

The methodological approach for this study was selected based on established spatial and epidemiological works that highlight the necessity of accounting for geographic clustering. Standard regression models often fail to capture the complex spatial dynamics inherent in health data. In selecting these foundational studies, priority was given to the relevance and robustness of their spatial methodologies rather than their specific years of publication. For instance, in evaluating spatial accessibility within Canadian urban areas, Shah et al. (2016) emphasize the

importance of employing local indicators of spatial association, such as Anselin's LISA and local Moran's I, to statistically confirm and identify spatial clusters. Furthermore, mapping techniques like bivariate local Moran maps have been successfully utilized by Ngom et al. (2016) to visualize morbidity "hot spots" in relation to the proximity of sports facilities. Because of these documented spatial behaviors, if spatial dependence is present, standard Ordinary Least Squares (OLS) regression may produce biased or inefficient estimates, making spatial regression models such as the Spatial Lag Model (SLM) or Spatial Error Model (SEM) more appropriate for capturing these spatial dynamics.

5. EXPECTED RESULTS & LIMITATIONS

The study expects to observe a negative association between recreational facility density and diabetes prevalence, where areas with greater access to recreational infrastructure tend to exhibit lower diabetes rates. This hypothesis is strongly supported by systematic reviews, such as that by De la Fuente et al. (2021), demonstrating that exposure to spaces equipped with sports amenities plays a protective role against Type 2 Diabetes by stimulating physical activity. Spatial analyses in Canadian contexts further corroborate this with Ngom et al. (2016) finding that populations living further away from spaces equipped with sports facilities experience a significantly higher prevalence of diabetes. However, several limitations should be acknowledged. First, the analysis relies on aggregated spatial units, which introduces the possibility of the Modifiable Areal Unit Problem (MAUP), meaning results may vary depending on how geographic boundaries are defined. Spatial accessibility literature confirms that the size and shape of the chosen unit are critical, as inconsistently sized administrative boundaries can lead to methodological inaccuracies such as the overestimation or underestimation of local access (Shah et al., 2016).

Second, recreational facility density serves only as a proxy for opportunities for physical activity and does not account for factors such as accessibility, affordability, quality of facilities, or whether individuals actually use them. It is a widely recognized limitation in environmental health research that spatial access to recreational parks does not inherently equate to actual utilization (Potestio et al., 2009), and studies frequently cite the absence of actual usage data as a primary constraint (Ngom et al., 2016). More importantly, diabetes prevalence is influenced by a wide range of determinants beyond physical activity, including diet, socioeconomic status, healthcare access, age distribution, and genetic predisposition. Ngom et al. (2016) note that demographic elements and lifestyle factors likely contribute significantly more to variations in diabetes rates than the simple presence or proximity of recreational infrastructure. As a result, the presence of recreational infrastructure alone cannot fully explain spatial variations in diabetes rates. Finally, because the analysis is cross-sectional and observational, causal relationships cannot be established; any observed relationships should therefore be interpreted as correlational rather than definitive evidence that recreational facilities directly reduce diabetes prevalence, which is repeatedly noted in similar cross-sectional spatial health studies (Potestio et al., 2009; De la Fuente et al., 2021)

References

1. De la Fuente, F., Saldías, M. A., Cubillos, C., Mery, G., Carvajal, D., Bowen, M., & Bertoglia, M. P. (2021). Green Space Exposure Association with Type 2 Diabetes Mellitus, Physical Activity, and Obesity: A Systematic Review. *International Journal of Environmental Research and Public Health*, 18(1), 97. https://doi.org/10.3390/ijerph18010097
2. Ngom, R., Gosselin, P., Blais, C., & Rochette, L. (2016). Type and Proximity of Green Spaces Are Important for Preventing Cardiovascular Morbidity and Diabetes—A Cross-Sectional Study for Quebec, Canada. *International Journal of Environmental Research and Public Health*, 13(4), 423. https://doi.org/10.3390/ijerph13040423
3. Potestio, M. L., Patel, A. B., Powell, C. D., McNeil, D. A., Jacobson, R. D., & McLaren, L. (2009). Is there an association between spatial access to parks/green space and childhood overweight/obesity in Calgary, Canada?. *International Journal of Behavioral Nutrition and Physical Activity*, 6(77). https://doi.org/10.1186/1479-5868-6-77
4. Shah, T. I., Bell, S., & Wilson, K. (2016). Spatial Accessibility to Health Care Services: Identifying under-Serviced Neighbourhoods in Canadian Urban Areas. *PLoS ONE*, 11(12): e0168208. https://doi.org/10.1371/journal.pone.0168208